

Moosa Irfaan

Graduate Student at New York University

moosairfaan0325@outlook.com • www.linkedin.com/in/moosa-irfaan • +1 (631) 992-5552 • <https://moosairfaan.dev/>

EDUCATION

NEW YORK UNIVERSITY

Masters of Science in Computer Engineering

ST. JOSEPH'S UNIVERSITY, NEW YORK

Bachelors of Science in Computer Science and Mathematics

- GPA: 3.7 / 4.0

New York, NY

SEPT 2026 - DEC 2027

Patchogue, NY

SEPT 2022 - MAY 2026

TECHNICAL HIGHLIGHTS

Full-stack development: React, TypeScript, Vite, FastAPI, RESTful APIs, PostgreSQL, Redis, SQLite, Railway, Vercel, Server-Sent Events (SSE)

Machine learning & AI: XGBoost, scikit-learn, CalibratedClassifierCV, ColumnTransformer, model calibration, real-time inference pipelines, faster-whisper transcription, LLM-based semantic scoring (Claude API), MediaPipe Face Landmarker, computer vision

Data engineering: ETL pipeline development, XBRL parsing, idempotent ingestion, Redis TTL deduplication, real-time web scraping, Playwright, schema normalization, SQLite optimization

Systems & infrastructure: Chrome Extension (Manifest V3), MutationObserver, event-driven architecture, notification systems, Twilio SMS, 99.9% uptime, GitHub Actions / CI-CD workflows, asynchronous processing, low-latency audio/video pipelines, macOS automation, PyObjC (Cocoa APIs)

PROJECTS

PosePulse (moggingML) — Computer Vision Assistant

Python • MediaPipe • Computer Vision

- Built a macOS computer vision application that detects user gaze in Photo Booth using MediaPipe Face Landmarker and automatically triggers synchronized audio playback and floating UI overlays — engineered a real-time pose state machine with temporal smoothing and grace periods to reduce false triggers, integrated Cocoa APIs (PyObjC) for active application detection, and optimized seamless audio looping with crossfades, fade transitions, and global hotkey controls for a responsive user experience.

Autocutter — AI-Powered Video Rough-Cut Tool

Python • FastAPI • React • Claude API

- Built a full-stack AI video editing pipeline that transcribes and scores footage to auto-generate a rough cut, cutting personal editing time from 6+ hours to under 2 hours — combines faster-whisper transcription, LLM-based segment scoring (Claude API) against target length/theme, and a constraint-based selection algorithm, with dual export to Final Cut Pro (FCPXML) or a custom React drag-and-trim timeline editor.

EDGAR Financial Dashboard - Real-Time SEC Filing Pipeline

Python • FastAPI • SQLite • React • XBRL

- Developed a full-stack data engineering pipeline ingesting SEC EDGAR XBRL 10-K filings into a RESTful FastAPI backend with SQLite — implemented idempotent ETL ingestion via INSERT OR IGNORE constraints, engineered a multi-key schema resolver handling inconsistent third-party API responses, and deployed to cloud infrastructure on Railway with automated startup ingestion.

Appsmith - Open Source Contribution (CI/CD Workflow Fix)

GitHub Actions • JavaScript • CI/CD

- Debugged a control-flow defect in Appsmith's (40K+ GitHub stars) GitHub Actions release-labeling workflow — traced a misplaced break statement causing premature loop termination that silently dropped QA labels on multi-issue pull requests, authored a fix with local test reproduction, and shipped it through the project's PR review pipeline with a 100% automated pre-merge check pass rate.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, Dart, C++, SQL, HTML/CSS

Frameworks & Libraries: React, FastAPI, Node.js, Flutter, scikit-learn, XGBoost, Recharts

Tools & Infrastructure: Git, GitHub Actions, Vite, Playwright, Redis, PostgreSQL, SQLite, SQLAlchemy, ffmpeg, dnd-kit

PROFESSIONAL EXPERIENCE

PEOPLETIME

Software Engineer Intern

Remote

JAN 2025 - JUNE 2025

- Developed cross-platform mobile application features using Flutter with backend services in Python and Java.
- Improved application load performance by 44% through state management and rendering optimizations.

FRED SILVER & COMPANY

Web Developer

Ronkonkoma, NY

NOV 2024 - JAN 2025

- Built an internal inventory management platform that automated manual Excel-based workflows for business operations.
- Optimized application performance through caching strategies and efficient data processing pipelines.